Introduction to Matrix

A matrix is a 2-dimensional array, which is a structured collection of numbers arranged in rows and columns. It is rectangular in form, providing a systematic way to organize and navigate through the data. For instance, a matrix A can be represented as:

$$A = \begin{matrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{matrix}$$

Where each number is an element identified by its positional coordinates, designated by its row and column.

Matrix Representation

A matrix is most commonly represented as a two-dimensional array in programming. Essentially, this means that a matrix is depicted as an array of arrays, where each sub-array represents a row containing multiple column elements. Thus, the primary array holds all the rows, making it possible to visualize the data in a grid-like structure reminiscent of a mathematical matrix.

For example, the above mentioned matrix is representation in Python as:

```
A = [[1, 2, 3],
     [4, 5, 6],
     [7, 8, 9]]
```

Accessing and Modifying Matrix Elements

In programming, elements of a matrix are accessed using their row and column indices. For instance, to access the element `5` in the matrix `A` mentioned above, one would reference its position using `A[1][1]`, considering a zero-based index system prevalent in most programming languages. Similarly, modifications to elements are performed by assigning a new value using the respective indices.

Example in Python:

```
# Accessing
print(A[1][1])  # Output: 5

# Modifying
A[1][1] = 10
print(A[1][1])  # Output: 10
```

Let's solve some coding problems to see matrix data structure in action.

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Problem 1: Richest Customer Wealth (Easy)

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